

Associations of maternal long chain polyunsaturated fatty acids, methyl mercury, and infant development in the Seychelles Child Development Nutrition Study

Fish consumption during gestation can provide the fetus with long chain polyunsaturated fatty acids (LCPUFA) and other nutrients essential for growth and development of the brain. However, fish consumption also exposes the fetus to the neurotoxicant, methyl mercury (MeHg). We studied the association between these fetal exposures and early child development in the Seychelles Child Development Nutrition Study (SCDNS). Specifically, we examined a priori models of Ω -3 and Ω -6 LCPUFA measures in maternal serum to test the hypothesis that these LCPUFA families before or after adjusting for prenatal MeHg exposure would reveal associations with child development assessed by the BSID-II at ages 9 and 30 months. There were 229 children with complete outcome and covariate data available for analysis. At 9 months, the PDI was positively associated with total Ω -3 LCPUFA and negatively associated with the ratio of Ω -6/ Ω -3 LCPUFA. These associations were stronger in models adjusted for prenatal MeHg exposure. Secondary models suggested that the MeHg effect at 9 months varied by the ratio of Ω -6/ Ω -3 LCPUFA. There were no significant associations between LCPUFA measures and the PDI at 30 months. There were significant adverse associations, however, between prenatal MeHg and the 30 month PDI when the LCPUFA measures were included in the regression analysis. The BSID-II Mental Developmental Index (MDI) was not associated with any exposure variable. These data support the potential importance to child development of prenatal availability of Ω -3 LCPUFA present in fish and of LCPUFA in the overall diet. Furthermore, they indicate that the beneficial effects of LCPUFA can obscure the determination of adverse effects of prenatal MeHg exposure in longitudinal observational studies.